

Show all of your work in the space provided to you. Full credit will not be awarded for correct answers unless sufficient supporting work is presented.

1. (20 pts) Consider the system of equations
- $$\begin{aligned} x + 5z &= 2 \\ -2x + y - 6z &= -1 \\ 2y + 8z &= 6 \end{aligned}$$

a) Write the system of equations as a **vector** equation.

$$x \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} + z \begin{bmatrix} 5 \\ -6 \\ 8 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$$

b) Write the system of equations as a **matrix** equation.

$$\begin{bmatrix} 1 & 0 & 5 \\ -2 & 1 & -6 \\ 0 & 2 & 8 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 6 \end{bmatrix}$$

c) Rewrite the augmented matrix in **echelon form** (does not need to be reduced echelon form). Show steps in the reduction.

$$\left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ -2 & 1 & -6 & -1 \\ 0 & 2 & 8 & 6 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 2 & 8 & 6 \end{array} \right] \sim \left[\begin{array}{ccc|c} 1 & 0 & 5 & 2 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

d) What does this tell you about the **type** of solution for the system of equations?
(You do not need to solve the system.)

Infinite solution set since the system is consistent and has a free variable

2. (12 pts) Let A be a 6×5 matrix. For each statement decide if it is always true, it is sometimes true and sometimes false, or it is always false. **No explanations required.**

- a) The columns of A span $\mathbb{R}^6$. Always true Always false Could be true or false
- b) The system $Ax = 0$ is consistent. Always true Always false Could be true or false
- c) The columns of A are linearly independent. Always true Always false Could be true or false
- d) The linear transformation $T(x) = Ax$ takes vectors in $\mathbb{R}^5$ and produces vectors in $\mathbb{R}^6$. Always true Always false Could be true or false

3. (20 pts) Given that the matrix A is row equivalent to

$$\begin{array}{c|cccccc} & B & B & F & B & F & B \\ \hline 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & -2 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{array}$$

a) Write the solution of $Ax = 0$ in parametric vector form.

$$x_1 = -x_3 - x_5$$

$$x_2 = 2x_3 - 2x_5$$

$$x_3 = x_3$$

$$x_4 = 3x_5$$

$$x_5 = x_5$$

$$x_6 = 0x_3 + 0x_5$$

$$\vec{x} = x_3 \begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ -2 \\ 0 \\ 3 \\ 1 \\ 0 \end{bmatrix}$$

where x_3 and x_5 can be any real numbers.

b) Describe the solution set geometrically.

This is a plane through the origin in $\mathbb{R}^6$ spanned by the vectors $\begin{bmatrix} -1 \\ 2 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -2 \\ 0 \\ 3 \\ 1 \\ 0 \end{bmatrix}$.

c) Is the system $Ax = b$ consistent for every b in $\mathbb{R}^6$? Explain.

Yes. There is a pivot in every row of $\text{RREF}(A)$ so you will never get a pivot in the augmented column.

d) Are there any vectors b for which $Ax = b$ has a unique solution? Explain.

No. There are 2 free variables so the solution set will always be infinite.

4. (9 pts) If $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ is a linear transformation with $T(\mathbf{e}_1) = (1, 2, 3)$ and $T(\mathbf{e}_2) = (1, 1, 1)$

a) What is $T(3, 10)$? $T(3, 10) = T(3\vec{e}_1 + 10\vec{e}_2) = 3T(\vec{e}_1) + 10T(\vec{e}_2)$
 $= 3(1, 2, 3) + 10(1, 1, 1) = (13, 16, 19)$

b) What is $T(x, y)$?

$$T(x, y) = x(1, 2, 3) + y(1, 1, 1) = (x+y, 2x+y, 3x+y)$$

c) What is the standard matrix for T ?

$$A = [T(\vec{e}_1) \quad T(\vec{e}_2)] = \begin{bmatrix} 1 & 1 \\ 2 & 1 \\ 3 & 1 \end{bmatrix}$$

5. (9 pts) Let $A = \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 3 & 0 \end{bmatrix}$.

a) Write the second column of AB as a linear combination of the columns of A .

$$2 \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + 1 \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix} + 0 \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix}$$

b) What is B^T ?

$$B^T = \begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 0 \end{bmatrix}$$

c) What is the 2,3 entry of $B^T A$?

$$\begin{bmatrix} 1 & 0 & 3 \\ 2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 4 & 7 \\ 2 & 5 & 8 \\ 3 & 6 & 9 \end{bmatrix}$$

2,3 entry is $[2 \ 1 \ 0] \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} = 14 + 8 = \boxed{22}$

6. (4 pts) What are the two properties that every linear transformation, T , must satisfy?

T must preserve vector addition

$$T(\vec{u} + \vec{v}) = T(\vec{u}) + T(\vec{v}) \quad \text{for all vectors } \vec{u} \text{ and } \vec{v}.$$

T must preserve scalar multiplication

$$T(c\vec{u}) = cT(\vec{u}) \quad \text{for all scalars } c \text{ and vectors } \vec{u}.$$

7. (8 pts) Let T be a linear transformation from $\mathbb{R}^5$ to $\mathbb{R}^3$ with standard matrix A .

a) What must be the size of the matrix?

A must have 3 rows and 5 columns, so 3×5

b) Is it possible for the mapping, T , to be one-to-one? Explain your answer.

Not possible. Since there are only 3 rows there can be at most 3 pivots, so at most 3 pivot columns.

But there are 5 columns so there must be 2 free variables. Therefore T cannot be one-to-one.

8. (6 pts) Let T be a linear transformation from $\mathbb{R}^5$ to $\mathbb{R}^3$ with standard matrix A . Circle each letter beside any statement listed below that is equivalent to the statement "The range of T is all of $\mathbb{R}^3$."

☒ a) There is a pivot in every row of $\text{RREF}(A)$.

☐ b) There is a pivot in every column of $\text{RREF}(A)$.

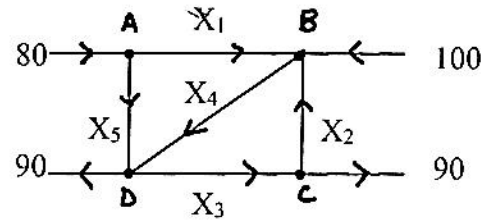
☒ c) The matrix equation $A\mathbf{x} = \mathbf{b}$ is consistent for every vector $\mathbf{b}$ in $\mathbb{R}^3$.

☐ d) The columns of A are linearly independent.

☒ e) $\mathbb{R}^3$ is equal to the span of the columns of A .

☒ f) Every vector in $\mathbb{R}^3$ can be written as a linear combination of the columns of A .

9. (12 pts) Consider the network shown here.



a) Write a system of linear equations that models the network flow.

Vertex A $\text{Flow}_{\text{in}} = \text{Flow}_{\text{out}}$
 $80 = x_1 + x_5$

Vertex B $x_1 + x_2 + 100 = x_4$

Vertex C $x_3 = x_2 + 90$

Vertex D $x_4 + x_5 = x_3 + 90$

$$\begin{aligned} x_1 + x_5 &= 80 \\ x_1 + x_2 - x_4 &= -100 \\ -x_2 + x_3 &= 90 \\ -x_3 + x_4 + x_5 &= 90 \end{aligned}$$

b) What is the reduced echelon form for the augmented matrix for this system?

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 1 & 80 \\ 1 & 1 & 0 & -1 & 0 & -100 \\ 0 & -1 & 1 & 0 & 0 & 90 \\ 0 & 0 & -1 & 1 & 0 & 90 \end{array} \right] \sim \left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 1 & 80 \\ 0 & 1 & 0 & -1 & -1 & -180 \\ 0 & 0 & 1 & -1 & -1 & -90 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]$$

c) If the flow along each edge must be nonnegative, find the maximum value possible for x_5 and the minimum value possible for x_4 ?

$$x_1 = 80 - x_5 \Rightarrow x_5 \leq 80 \text{ to make } x_1 \geq 0$$

$$x_2 = x_4 + x_5 - 180$$

$$x_3 = x_4 + x_5 - 90$$

If the max value for x_5 is 80 then
 x_4 must be at least 100 to make $x_2 \geq 0$

max value for x_5 is 80

min value for x_4 is 100

Note this also means that $x_4 + x_5$ must be at least 180
to make $x_2 \geq 0$ so x_3 will have to be at least 90
One solution to the system would be

$$x_1 = 0, x_2 = 0, x_3 = 90, x_4 = 100, x_5 = 80$$